

The Mental Health Crisis in Sports: The Perfect Storm of Contemporary Factors

Claudia L. Reardon, MD

Department of Psychiatry, University of Wisconsin School of Medicine and Public Health, Madison

LAYING THE GROUNDWORK

Athletes are susceptible to mental health symptoms and disorders because they are human and because of the risk factors imposed by sport.¹ The current mental health crisis is making participation in sport and life even more challenging for this population. The American Academy of Pediatrics, American Academy of Child and Adolescent Psychiatry, and others have declared a national emergency in child and adolescent mental health.² Suicide is now the second leading cause of death for young people (ages 10–34 years).³ Moreover, depression, anxiety, eating disorders, and substance use disorders among athletes are associated with early attrition from sport,⁴ a higher risk of injuries,⁵ negative effects on sports performance,⁶ burnout,⁷ and lower levels of academic and occupational functioning.⁶ Importantly, aspects of the crisis are unique and concerning in the world of sport. Physicians, other licensed mental health professionals, athletic trainers, coaches, and other members of the athlete entourage witness this painful reality every day in their offices and on the playing fields. This editorial highlights several factors that may be synergizing to create the perfect storm leading to this crisis in the world of sport: emerging adulthood as a particularly challenging phase of life, discrimination and racism, increasing professionalization of sports at younger ages, a lack of mental health literacy and resources, the effect of technology and the 24/7 world, and the COVID-19 pandemic.

EMERGING ADULTHOOD AS A PARTICULARLY CHALLENGING PHASE OF LIFE

Emerging adulthood spans the years from the late teens to mid-20s, a time that overlaps with peak years of athletic achievement⁸ and with the peak incidence age for many mental health disorders.⁹ For most young people, this is a period of transition from the dependency of adolescence to the responsibility and independence of young adulthood. It involves developing a stable identity; learning to self-regulate; forming intimate relationships with others; learning skills or habits to live independently; moving out of the family home; obtaining higher education, preparing for a career, or both; and moving into the workforce. High-level sport involvement can be time consuming to the point that it alters these typical development timelines if insufficient time or opportunity is available to accomplish

the milestones. “Helicopter parents” have the best of intentions as they try to help their heavily scheduled athletes but may inadvertently contribute to young adults feeling incapable, vulnerable, and dependent.¹⁰ Young athletes who perceive parental overinvolvement in sport report worse mental health.¹¹ The COVID-19 pandemic exacerbated a lack of opportunities for young adults, including athletes, to accomplish the typical milestones of emerging adulthood.

Furthermore, emerging adults do not yet have the life experience to understand that, when confronted with a stressor, in many cases, “this too shall pass.” For example, recent data showed that humiliation and a sense of entrapment in current life circumstances are 2 particularly dire risk factors for mental health symptoms in athletes.¹² We also know that many athletes, at one time or another, experience these feelings.¹² Their immature brain developmental status and incomplete set of life experiences may not allow them to appreciate that the humiliation or sense of entrapment—whether within or outside of sport—may well not last forever.

DISCRIMINATION AND RACISM

In the United States and many other locations, an important determinant of health—including mental health—is ongoing systemic, institutional, interpersonal, and internalized racism and discrimination.¹³ Although nothing new, increasing attention is appropriately being drawn to this problem. Racism and discrimination contribute to the inequitable risk of poor mental health among minority athletes.^{14–16} Minority athletes often find that no available providers look like them or identify as they do, and this can deter help-seeking.¹⁷ At the same time, they may feel the emotional weight of being seen as a representative for an entire community or needing to win for their entire race or culture.¹⁸ They may be told to “just shut up and play,” as any references they make to social injustices (eg, via notations on their uniforms or behaviors during sporting events) are critically scrutinized.¹⁹ Ultimately, racism, discrimination, and disparate access to resources have a bearing on all of the other contributing factors in the perfect storm of the mental health crisis. For example, young athletes from lower socioeconomic groups have suffered from even greater increases in rates of depression than other socioeconomic groups during the COVID-19 pandemic.²⁰

INCREASING PROFESSIONALIZATION OF SPORTS AT YOUNGER AGES

Youth athletes today are often required to put as much time into their sport as collegiate athletes historically have been.²¹ For collegiate athletes, sport is at least the equivalent of a full-time job.²² Parents receive pressure from many angles to sign their children up for sport and to do whatever it takes, including hiring individual coaches and promoting year-round participation in a single sport from a very young age, to ensure success in sport.²³ They may sense that their children will miss out on important life opportunities if they do not do these things and that endless practice is the route to athletic mastery. This mindset may lend itself to coaching styles that are performance centered (focused on demonstrating superior performance compared with others) rather than skills mastery centered (focused on exerting effort and improving personally in specific tasks), with the former approach associated with worse mental health and higher rates of burnout in athletes.²⁴ The end result of all of this may be higher rates of injuries (which themselves are associated with more mental health disorders)⁶; poor life balance; insufficient time for school, friends, family, and sleep; a crushing sense of pressure²¹; and perversely, less lifelong sport involvement.²⁵ How can young people exit from sport, if they decide that it is not what they want, when they know how much time, effort, and resources have been poured into their participation? Have they had a chance to develop their identity outside of sport, and if not, what do they have left if and once they do exit from sport?

LACK OF MENTAL HEALTH LITERACY AND RESOURCES

Mental health literacy is defined as the knowledge, beliefs, and skills regarding mental health disorders that aid their recognition, management, or prevention.²⁶ It includes concepts related to the knowledge of effective self-management strategies, challenging mental disorder stigma, the awareness of and ability to assist others who are struggling, and the facilitation of help-seeking behaviors.²⁶ The general public and particularly athletes (who often exist in a problematic culture of toughness) frequently have low rates of mental health literacy.²⁶ This makes it difficult for athletes to reach out for help or even to be able to label their experiences as mental health symptoms when appropriate. Similarly, if their coaches and other support staff have not been educated in mental health symptoms and disorders, how are staff to even know what constitutes negative, stigmatizing, help-dissuading commentary that should be avoided? Of course, mental health resource limitations¹ and organizational practices may constrain the ability of coaches to fully support mental health in word and deed.²⁶

EFFECT OF TECHNOLOGY AND THE 24/7 WORLD

Young people have markedly increased access to information and technology compared with generations past. For athletes, this is a double-edged sword.²⁷ It may help with opportunities for sponsorships and an ability to make a living. However, they are vulnerable to constant criticism, harassment, and abuse from sports fans who feel

relatively protected by the pseudoanonymity of an online presence. Even the seemingly nicest of people do not hesitate to criticize athletes—who apparently are superhuman and should not be fazed by such words—in public forums. Athletes may worry that they need to constantly check their online media profiles to maintain engagement with their fan base, so it can be challenging to avoid the abusive or critical commentary.²⁷

Moreover, with regard to the big picture and when mental health really plummeted for young people, many have traced it to the widespread expansion of the use of smartphones.²⁸ Illustrating this timeline, between 2008 and 2017, the percentage of people experiencing serious psychological distress in the prior month increased by 71% in the 18- to 25-year-old age group.²⁹ The pandemic only hastened our move to an all-virtual world. Head-down use of phones has replaced meaningful face-to-face interactions and other activities more beneficial for mental health. A constant barrage of digitally altered social media posts leads athletes—who already disproportionately suffer from eating and other body image-related disorders³⁰—to a place of anxiogenic comparison with others.³¹ Additionally, the more screen time adolescents and young adults have, the less sleep they get,²⁹ and sleep is particularly important in athletes for the purposes of recovery and avoiding injury.³²

THE COVID-19 PANDEMIC

The COVID-19 pandemic has gutted people's sense of safety, stability, and routine. Athletes have suffered substantially during this time. After the cancellation of organized sports in spring 2020, the negative mental health effects were immediately obvious. A full 40% of 13 000 adolescent US athletes reported moderate to severe symptoms of depression at the time.²⁰ One state's data showed a jump in adolescent athlete depression rates from 10% before the pandemic to 33% shortly after it started.³³ Perhaps more alarming is that, despite the resumption of sport and other aspects of normal life, prepandemic rates of depression in athletes have not reverted to baseline.³⁴ Athletes are still catching up from the years of lost social support from teammates and coaches, the loss of the antidepressant and anxiolytic benefits of sport and exercise, the academic losses for those who are student-athletes, and reduced access to support services.³⁴ Furthermore, many athletes—whether competitive high school or college seniors when the pandemic struck or elite athletes nearing the end of their competitive careers—unexpectedly faced premature retirement from competitive sport. It is well-established that unanticipated, unplanned transition out of sport, without time to contemplate and adjust to that new reality, confers a significant risk for mental health symptoms,³⁵ and that is precisely what the pandemic wrought.

TOWARD A BRIGHTER FUTURE

Mental health in sport is receiving unprecedented attention in medical and research circles and among the general public.¹ This is hopeful. Mental health stigma is alive and well but is showing evidence of lessening, at least in some places. High-level athletes are courageously sharing their experiences. Society needs to continue to move toward seeing mental health as inseparable from other

aspects of physical health. Health is health. The brain is a real organ, just like the heart and lungs, and athletes will not be at their best in sport or in life if the brain is unwell. Stakeholders need to focus on improving all the risk factors discussed earlier in the current crisis, and simultaneously, they cannot forget about the other end of the mental health continuum: we must optimize the protective factors that provide a buffer against mental health problems. All stakeholders must work together toward a culture of mental health literacy, cultural competency, and positive support in the sporting environment.³⁶ Specific actions stakeholders can take include the following:

- Young athletes should be supported while being allowed to make mistakes and learn from them. The focus should remain on process goals rather than outcomes as much as possible.²⁴ All athletes should receive the consistent message that their worth does not depend on their success in sport.
- Clinicians providing mental health care to athletes should be licensed health care providers who receive training related to trauma-informed care, which recognizes the presence of trauma symptoms, acknowledges the role trauma may play in an individual's life, and avoids practices that might exacerbate trauma symptoms. Care should also be culturally competent, with appreciation for the social determinants of health.³⁷
- Intense training in a single sport to the exclusion of others should be delayed until at least late adolescence, if not longer.³⁸
- Educational programming targeting mental health literacy should be offered to athletes, caregivers, coaches, and personnel all the way up to the highest levels of sports organizations, and it should be paired with practical interventions where mental health treatment referral options are sparse.²⁶
- Athletes should receive education on the effect of media use on their mood, anxiety, and sleep and strategies for minimizing negative effects on these factors.¹⁸
- Policymakers and organizations should consider the mental health effects of decisions related to COVID-19 or any other systemic threats to sports participation.²⁰

Stakeholders in sport must appreciate the gravity of the current mental health crisis. They have an ethical obligation to take these and other steps to safeguard athletes' health, well-being, and ability to function in sport and in life.

REFERENCES

1. Reardon CL, Hainline B, Aron CM, et al. Mental health in elite athletes: International Olympic Committee consensus statement. *Br J Sports Med.* 2019;53(11):667–699. doi:10.1136/bjsports-2019-100715
2. AAP-AACAP-CHA Declaration of a National Emergency in Child and Adolescent Mental Health. American Academy of Pediatrics. Updated October 19, 2021. Accessed May 8, 2023. <https://www.aap.org/en/advocacy/child-and-adolescent-healthy-mental-development/aap-aacap-cha-declaration-of-a-national-emergency-in-child-and-adolescent-mental-health/>
3. Underlying cause of death. Centers for Disease Control and Prevention. Reviewed January 11, 2023. Accessed May 8, 2023. <https://wonder.cdc.gov/Deaths-by-Underlying-Cause.html>
4. Vella SA, Clifff DP, Magee CA, Okely AD. Associations between sports participation and psychological difficulties during childhood: a two-year follow up. *J Sci Med Sport.* 2015;18(3):304–309. doi:10.1016/j.jsams.2014.05.006
5. Putukian M. The psychological response to injury in student athletes: a narrative review with a focus on mental health. *Br J Sports Med.* 2016;50(3):145–148. doi:10.1136/bjsports-2015-095586
6. Chang CJ, Putukian M, Aerni G, et al. Mental health issues and psychological factors in athletes: detection, management, effect on performance, and prevention: American Medical Society for Sports Medicine Position Statement. *Clin J Sports Med.* 2020;30(2):e61–e87. doi:10.1097/JSM.0000000000000817
7. Gould D. Intensive sport participation and the prepubescent athlete: competitive stress and burnout. In: Cahill BR, Pearl AJ, eds. *Intensive Participation in Children's Sport.* Human Kinetics; 1993:19–38.
8. Allen SV, Hopkins WG. Age of peak competitive performance of elite athletes: a systematic review. *Sports Med.* 2015;45(10):1431–1441. doi:10.1007/s40279-015-0354-3
9. Kessler RC, Berglund P, Demler O, Jin R, Merikangas KR, Walters EE. Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication. *Arch Gen Psychiatry.* 2005;62(6):593–602. doi:10.1001/archpsyc.62.6.593
10. Wieland DM, Kucirka BG. Helicopter parenting and the mental health of iGen college students. *J Psychosoc Nurs Ment Health Serv.* 2020;58(5):16–22. doi:10.3928/02793695-20191210-01
11. Salla J, Michel G. Parental sport over-involvement and anxiety among youth tennis athletes. *Sci Sports.* 2014;29:S40. doi:10.1016/j.scispo.2014.08.079
12. McLoughlin E, Fletcher D, Slavich GM, Arnold R, Moore LJ. Cumulative lifetime stress exposure, depression, anxiety, and well-being in elite athletes: a mixed-method study. *Psychol Sport Exerc.* 2021;52:101823. doi:10.1016/j.psychsport.2020.101823
13. Lipson SK, Kern A, Eisenberg D, Breland-Noble AM. Mental health disparities among college students of color. *J Adolesc Health.* 2018;63(3):348–356. doi:10.1016/j.jadohealth.2018.04.014
14. Kroshus E, Davoren AK. Mental health and substance use of sexual minority college athletes. *J Am Coll Health.* 2016;64(5):371–379. doi:10.1080/07448481.2016.1158179
15. Ballesteros J, Tran AGTT. Under the face mask: racial-ethnic minority student-athletes and mental health use. *J Am Coll Health.* 2020;68(2):169–175. doi:10.1080/07448481.2018.1536663
16. Tran AGTT. Looking forward to student-athlete mental health: racial/ethnic trends from 2010 to 2015. *J Am Coll Health.* 2021;69(8):942–950. doi:10.1080/07448481.2020.1725018
17. Wilkerson TA, Stokowski S, Fridley A, Dittmore SW, Bell CA. Black football student-athletes' perceived barriers to seeking mental health services. *J Issues Intercolleg Athl.* 2020;55–81. College Sport Research Institute. Accessed May 8, 2023. http://csri-jiia.org/wp-content/uploads/2020/02/SI_2020_01_03.pdf
18. Faustin M, Burton M, Callender S, Watkins R, Chang C. Effect of media on the mental health of elite athletes. *Br J Sports Med.* 2022;56(3):123–124. doi:10.1136/bjsports-2021-105094
19. Platt L. Athlete activism is on the rise, but so is the backlash. Global Sport Matters. Published April 16, 2018. Accessed May 8, 2023. <https://globalsportmatters.com/culture/2018/04/16/athlete-activism-is-on-the-rise-but-so-is-the-backlash/>
20. McGuire TA, Biese KM, Petrovska L, et al. Mental health, physical activity, and quality of life of U.S. adolescent athletes during COVID-19-related school closures and sport cancellations: a study of 13 000 athletes. *J Athl Train.* 2021;56(1):11–19. doi:10.4085/1062-6050-0478.20
21. DI time management. National Collegiate Athletic Association. Accessed May 8, 2023. http://fs.ncaa.org/Docs/eligibility_center/Student_Resources/Time_Management_DI_DII_DIII.pdf
22. Brewer BW, Petitpas AJ. Athletic identity foreclosure. *Curr Opin Psychol.* 2017;16:118–122. doi:10.1016/j.copsyc.2017.05.004
23. Bell DR, Post EG, Trigsted SM, Hetzel S, McGuire TA, Brooks MA. Prevalence of sport specialization in high school athletics: a

- 1-year observational study. *Am J Sports Med.* 2016;44(6):1469–1474. doi:10.1177/0363546516629943
24. Kroshus E. Mind, body and sport: risk factors in the sport environment. National Collegiate Athletic Association. Accessed May 8, 2023. <https://www.ncaa.org/sports/2014/11/5/mind-body-and-sport-risk-factors-in-the-sport-environment.aspx>
 25. Sagas M. What does science say about athletic development in children? The Aspen Institute. Published September 2013. Accessed May 8, 2023. <https://www.aspeninstitute.org/wp-content/uploads/2016/06/Project-play-september-2013-roundtable-research-brief.pdf>
 26. Gorczynski P, Currie A, Gouttebargue V, et al. Developing mental health literacy and cultural competence in elite sport. *J Appl Sport Psychol.* 2021;33(4):387–401. doi:10.1080/10413200.2020.1720045
 27. Pocock M, Skey M. ‘You feel a need to inspire and be active on these sites otherwise...people won’t remember your name’: elite athletes and the need to maintain ‘appropriate distance’ navigating online gendered space. *New Media Soc.* Published online January 27, 2022. doi:10.1177/14614448211069343
 28. Twenge JM. Why increases in adolescent depression may be linked to the technological environment. *Curr Opin Psychol.* 2020;32:89–94. doi:10.1016/j.copsyc.2019.06.036
 29. Twenge JM, Cooper AB, Joiner TE, Duffy ME, Binau SG. Age, period, and cohort trends in mood disorder indicators and suicide-related outcomes in a nationally representative dataset, 2005–2017. *J Abnorm Psychol.* 2019;128(3):185–199. doi:10.1037/abn0000410
 30. Wells KR, Jeacocke NA, Appaneal R, et al. The Australian Institute of Sport (AIS) and National Eating Disorders Collaboration (NEDC) position statement on disordered eating in high performance sport. *Br J Sports Med.* 2020;54(21):1247–1258. doi:10.1136/bjsports-2019-101813
 31. Encel K, Mesagno C, Brown H. Facebook use and its relationship with sport anxiety. *J Sports Sci.* 2017;35(8):756–761. doi:10.1080/02640414.2016.1186817
 32. Kroshus E, Wagner J, Wyrick D, et al. Wake up call for collegiate athlete sleep: narrative review and consensus recommendations from the NCAA Interassociation Task Force on Sleep and Wellness. *Br J Sports Med.* 2019;53(12):731–736. doi:10.1136/bjsports-2019-100590
 33. McGuine TA, Biese KM, Petrovska L, et al. Changes in the health of adolescent athletes: a comparison of measures collected before and during the COVID-19 pandemic. *J Athl Train.* 2021;56(8):836–844. doi:10.4085/1062-6050-0739.20
 34. McGuine T, Biese K, Hetzel S, et al. A multi-year assessment of sport participation during the COVID-19 pandemic on the health of adolescent athletes. *J Athl Train.* 2023;58(1):44–50. doi:10.4085/1062-6050-0679.21
 35. Brown JC, Kerkhoffs G, Lambert MI, Gouttebargue V. Forced retirement from professional rugby union is associated with symptoms of distress. *Int J Sports Med.* 2017;38(8):582–587. doi:10.1055/s-0043-103959
 36. IOC Mental Health in Elite Athletes Toolkit. Spring 2021. International Olympic Committee. Accessed May 8, 2023. <https://stillmed.olympics.com/media/Document%20Library/IOC/Athletes/Safe-Sport-Initiatives/IOC-Mental-Health-In-Elite-Athletes-Toolkit-2021.pdf>
 37. Comas-Díaz L, Hall GN, Neville HA. Racial trauma: theory, research, and healing: introduction to the special issue. *Am Psychol.* 2019;74(1):1–5. doi:10.1037/amp0000442
 38. Jayanthi N, Pinkham C, Dugas L, Patrick B, Labella C. Sports specialization in young athletes. *Sports Health.* 2013;5(3):251–257. doi:10.1177/1941738112464626

Address correspondence to Claudia L. Reardon, MD, Department of Psychiatry, University of Wisconsin School of Medicine and Public Health, 6001 Research Park Blvd, Madison, WI 53719. Address email to clareardon@wisc.edu.